

Efficiency

Economic question

2) When do choices made in the pursuit of *self-interest* also promote *social interest*?

Self-interest - choices that you think are *best for you*

Social interest - choices that are *best for society*

> Think of your own individual interests

Willingness to pay and sell

- Consumers many times are willing to pay more than the market price.
 - A consumer is willing to purchase a good if the price is below their *maximum willingness to pay*.
- Producers likewise are willing to sell for less than the market price.
 - A producer is willing to sell a good if the price is above their *minimum willingness to sell*.

Marginal Benefit

- Marginal benefit is the benefit that a person receives from consuming one more unit of a good/service.
- Marginal benefit is measured as the maximum amount that a person is willing to pay for one more unit of a good/service.
- As quantity increases, the marginal benefit from consuming one additional unit of a good decreases – principle of decreasing marginal benefit.
- How much you value a good depends on your marginal benefit from consuming one additional unit of that good – it is not the same as the price you pay for the good

Marginal Benefit & Demand

- Value or marginal benefit depend on the willingness to pay for an additional unit of a good or a service which in turn determines the demand for a good/service.
- An individual's marginal benefit curve is an individual's demand curve.
- The market demand curve is the horizontal sum of the individual demand curves and is formed by adding the quantities demanded by all the individuals at a particular price.
- Market demand curve = marginal social benefit curve

Consumer Surplus

- We do not always have to pay what we are actually willing to pay
- When people buy something for less than it is worth to them, they receive a consumer surplus.
- Consumer Surplus: The benefit that a consumer gets from consuming a good above and beyond the actual price of the good.

Consumer Surplus

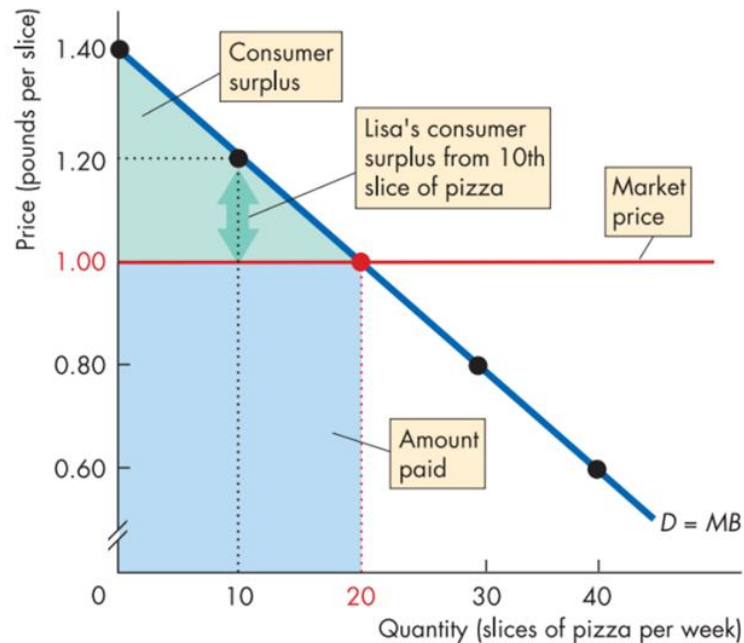
How do we calculate consumer surplus?

- By calculating the area below the demand curve and above the price level i.e. the area of the triangle between price, demand curve and the y-axis.
- Since the demand curve gives us the willingness to pay and the price line gives us what we actually have to pay for the good, the gap between them is essentially our consumer surplus.

Lisa's consumer surplus = area of green triangle

$$\begin{aligned} &= \frac{1}{2} \times 0.40 \times 20 \\ &= \text{£ } 4 \end{aligned}$$

A Consumer's Demand and Consumer Surplus



Marginal Cost

- Marginal cost is the cost of producing one more unit of a good/service.
- Cost is what the producer gives up – price is what the producer receives.
- Marginal cost is the minimum price that a producer must receive to produce or sell another unit of a good/service.
- If price is lower than the marginal cost of production, producers will not produce the good.

Marginal Cost & Supply

- Marginal cost is also the supply curve.
- The market supply curve is the horizontal sum of the individual supply curves and is formed by adding the quantities supplied at each price level.
- Market supply curve = marginal social cost curve

Producer Surplus

- If price received by producers is greater than the costs they incur, they earn a producer surplus.
- Producer Surplus: The benefit that a producer gets from producing a good at a cost below the actual price of the good.

Producer Surplus

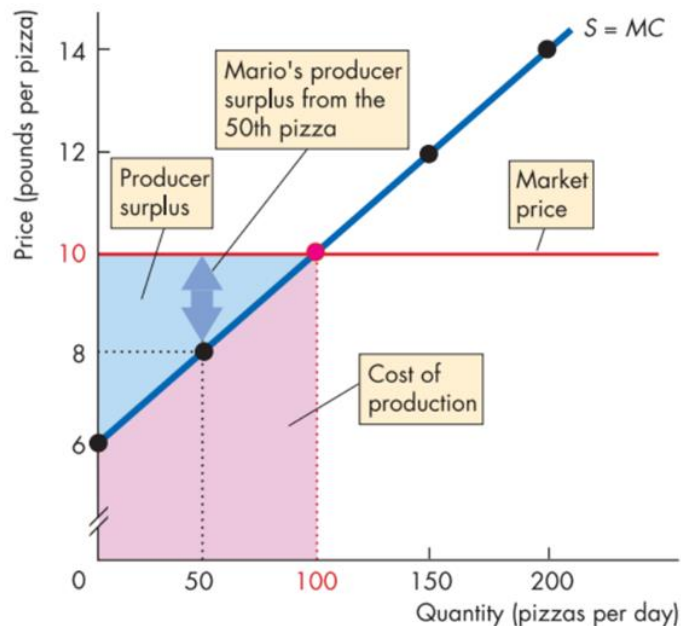
How do we calculate producer surplus?

- By calculating the area above the supply curve and below the price level i.e. the area of the triangle between price, supply curve and the y-axis.
- Since the supply curve gives us the marginal cost incurred by the firm and the price line gives us what producers actually get for the good, the gap between them is essentially the producer surplus.

Mario's producer surplus = area of blue triangle

$$\begin{aligned} &= \frac{1}{2} \times 100 \times 4 \\ &= \text{£ } 200 \end{aligned}$$

A Producer's Supply and Producer Surplus



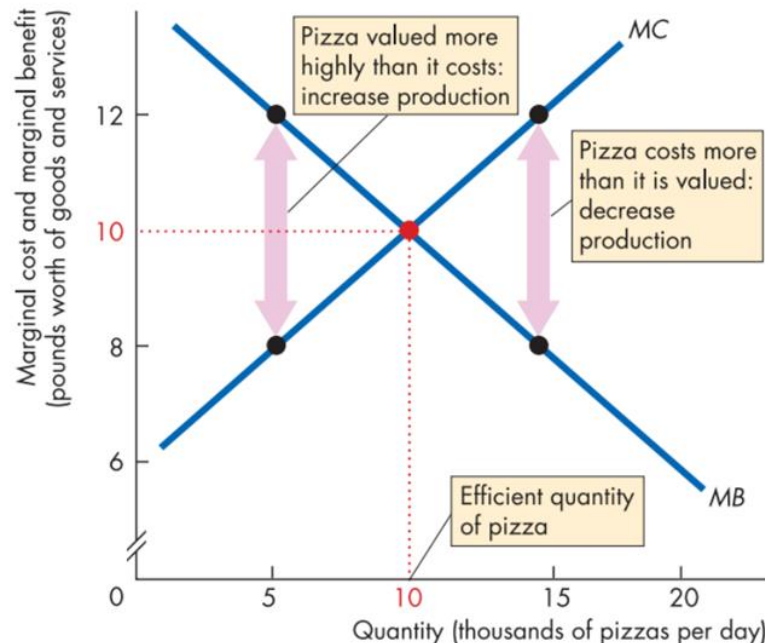
Measuring surplus

- When a consumer buys a good below the market price, this creates value.
 - Known as *consumer surplus*, a measure of consumers' benefit from the purchase.
- When a producer sells a good above the market price, this creates value.
 - Known as *producer surplus*, a measure of producers' benefit from the sale.
- *Surplus* is measured as the difference between the price at which at which a buyer or seller would be willing to trade and the actual price.

Market Efficiency

- Is the free-market equilibrium an efficient outcome?
- Is there any other outcome at which producers and consumers are better off?
- We know equilibrium occurs at a point where demand = supply
- In other words, at equilibrium $MB = MC$
- If marginal benefit > marginal cost, producers have an incentive to increase production.
- If marginal benefit < marginal cost, producers should decrease production

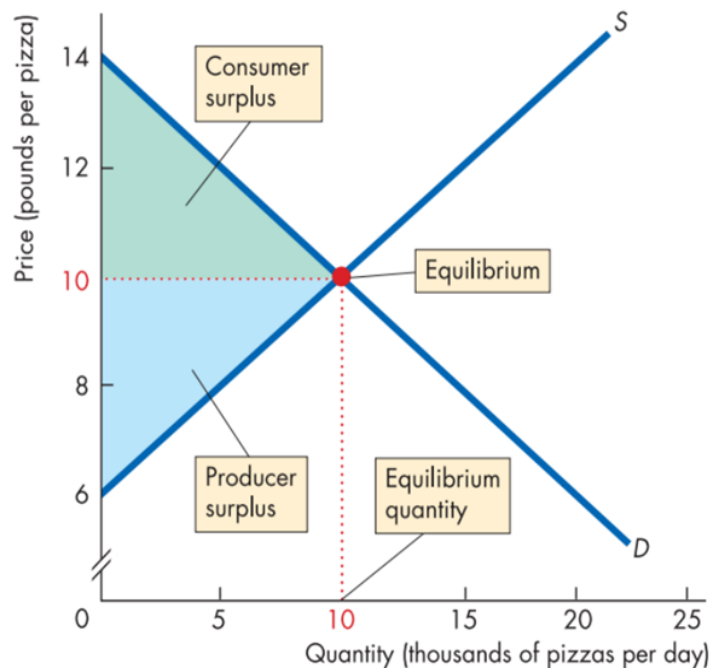
The Efficient Quantity of Pizza



Market Efficiency

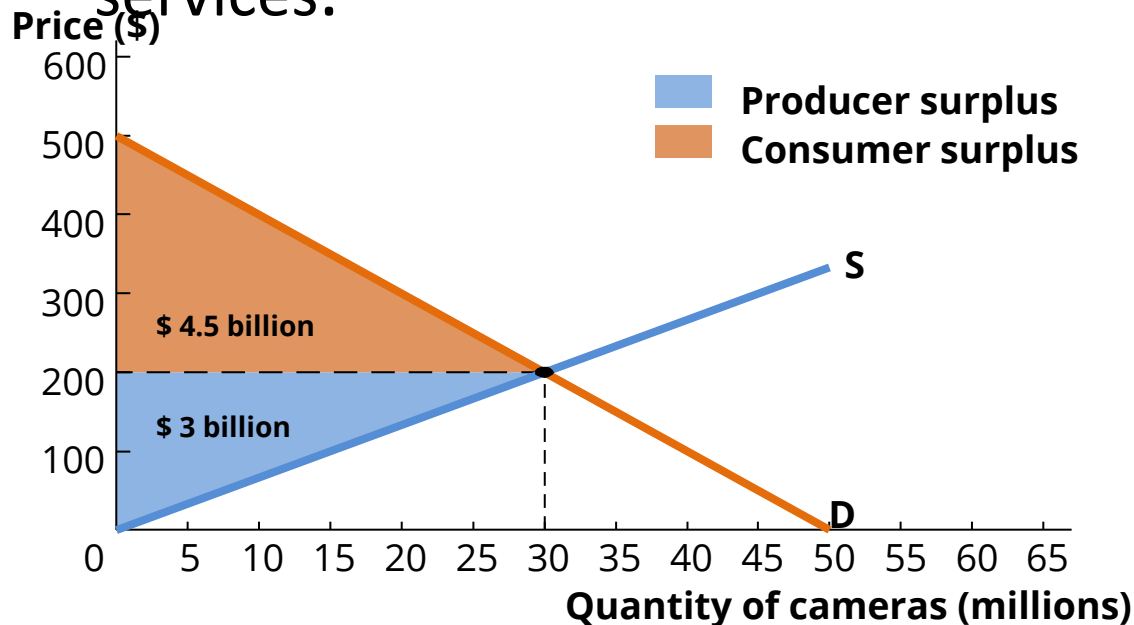
- Is the free-market equilibrium an efficient outcome?
- Another approach is to evaluate total surplus.
- The total surplus or economic surplus is the net benefit achieved from the transaction.
- **Total surplus = consumer surplus + producer surplus**
- Economic efficiency is achieved when total surplus is maximized.
- In other words, the competitive equilibrium where demand = supply is the point that maximizes social welfare.

An Efficient Market for Pizza



Total surplus

Total surplus is the combined benefits that everyone receives from participating in an exchange of goods or services.



- Total *consumer surplus* is equal to the area underneath the demand curve and above the equilibrium price.
– $CS = \frac{1}{2} \times 30M \times (\$500 - \$200) = \$4.5B$
- Total *producer surplus* is equal to the area above the supply curve and below the equilibrium price.
– $PS = \frac{1}{2} \times 30M \times (\$200 - \$0) = \$3B$
- *Total surplus* = $CS + PS$.

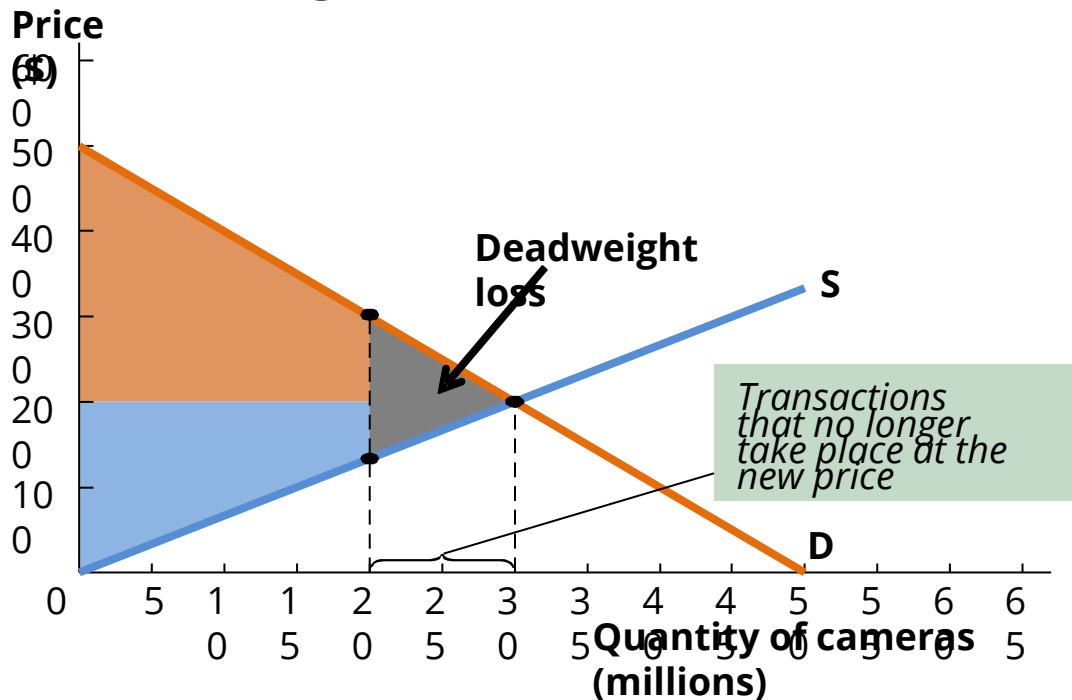
Market Failures & Deadweight Loss

- Sometimes markets fail to achieve an efficient outcome – this is called a market failure
- Market failure can occur because
 - Too little of a good is produced (underproduction)
 - Too much of a good is produced (overproduction)

Deadweight loss is the decrease in total surplus that results from producing an inefficient quantity of the good.

Deadweight loss

When an artificial price is imposed on a market, a deadweight loss occurs.



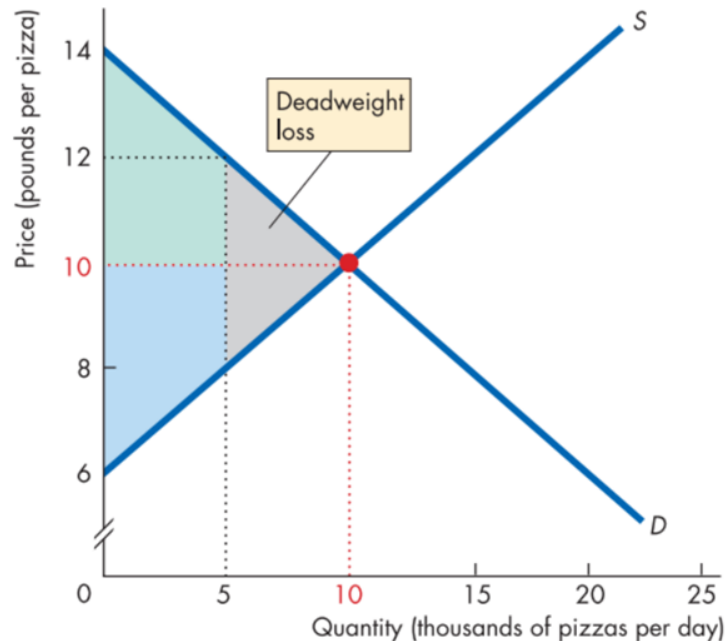
The *deadweight loss* is the loss of total surplus that results when the quantity of a good that is bought and sold is below the market equilibrium quantity.

- Reduction in cameras sold by 10 million.
- Reduces consumer and producer surplus.
- Deadweight loss is the gray triangle.

Market failures & Deadweight loss

Underproduction

- At equilibrium, 10,000 pizzas are produced in a day – the efficient outcome
- If pizza production is reduced to 5000 pizzas in a day, there is underproduction and a deadweight loss arises.
- The deadweight loss equals the loss in total surplus i.e. the grey area
- This loss is borne by the entire society – both consumers and producers

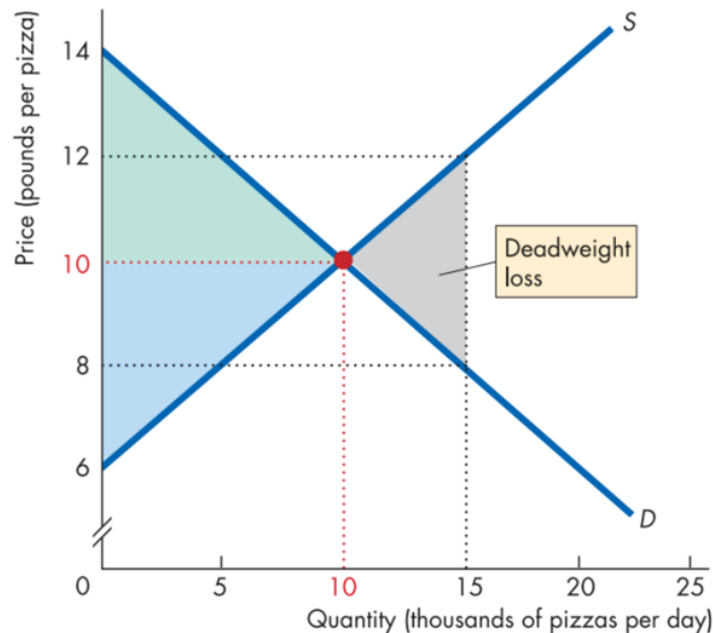


(a) Underproduction

Market failures & Deadweight loss

Overproduction

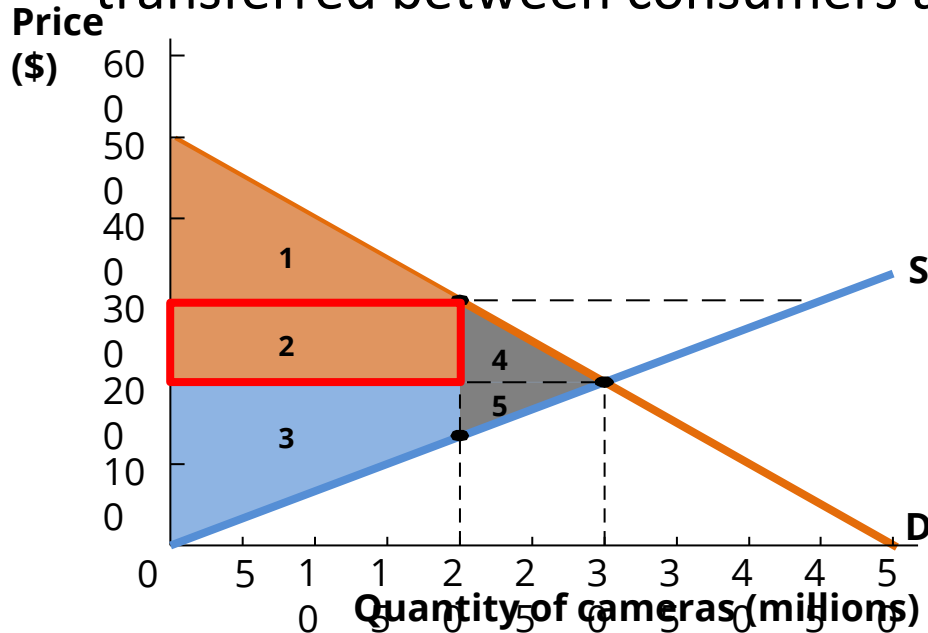
- At equilibrium, 10,000 pizzas are produced in a day – the efficient outcome
- If pizza production is increased to 15000 pizzas in a day, there is overproduction and a deadweight loss arises.
- At 15000 pizzas, consumers are only willing to pay £8 for an additional pizza but marginal cost equals £12.
- The deadweight loss equals the loss in total surplus i.e. the grey area



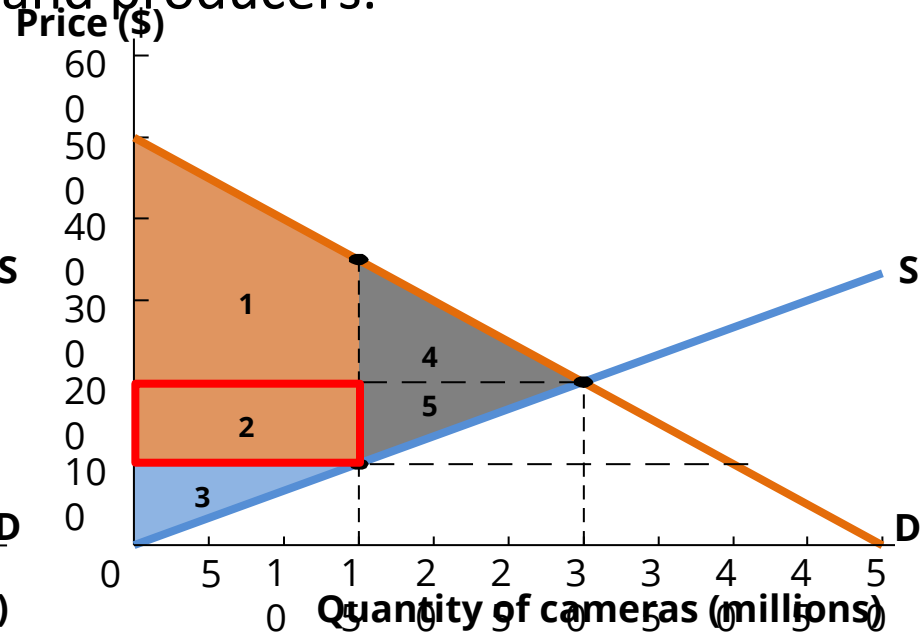
(b) Overproduction

Changing the distribution of total surplus

When an artificial price is imposed on a market, surplus is transferred between consumers and producers.



When the price is raised above \$200, consumer surplus area 2 is transferred to producers.



When the price is lowered below \$200, producer surplus area 2 is transferred to consumers.

Sources of Market Failures

Common reasons behind market failures (underproduction or overproduction) are:

- Price regulations
- Taxes and subsidies and quotas
- Externalities
- Public goods and common resources
- Monopoly

Price regulations

- *Price ceiling/price cap* - a government regulation that makes it illegal to charge a price higher than a specified level e.g. a rent ceiling
- *Price floor* - a government regulation that makes it illegal to charge/pay a price lower than a specified level e.g. minimum wage

What effect does a rent ceiling have on a housing market?

What about efficiency?

Because of the rent ceiling, there is underproduction in the market – an inefficient quantity of 4400 houses are traded

Consumer surplus shrinks

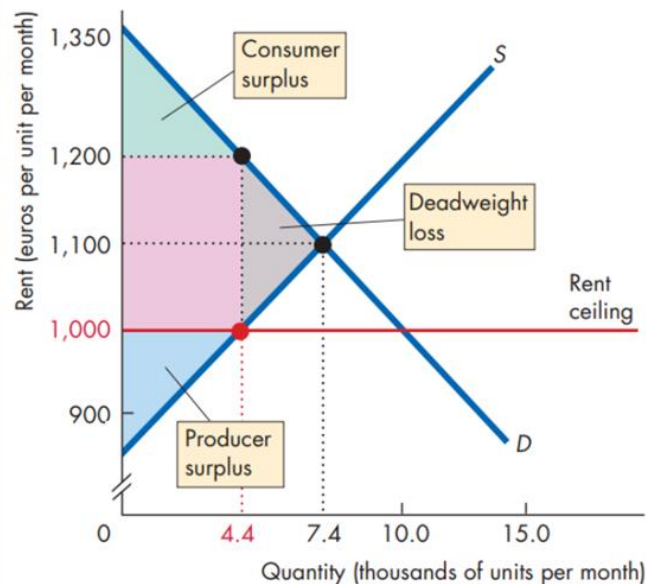
Producer surplus shrinks

Deadweight loss (grey triangle) arises

Increased search activity is represented by the red rectangle

Full loss from rent ceiling equals the deadweight loss (grey) plus the cost of increased search activity (red)

Figure 6.2 The Inefficiency of a Rent Ceiling



What effect does a minimum wage policy have on a labor market?

What about efficiency?

A minimum wage leads to an inefficient outcome

Firms' surplus shrinks

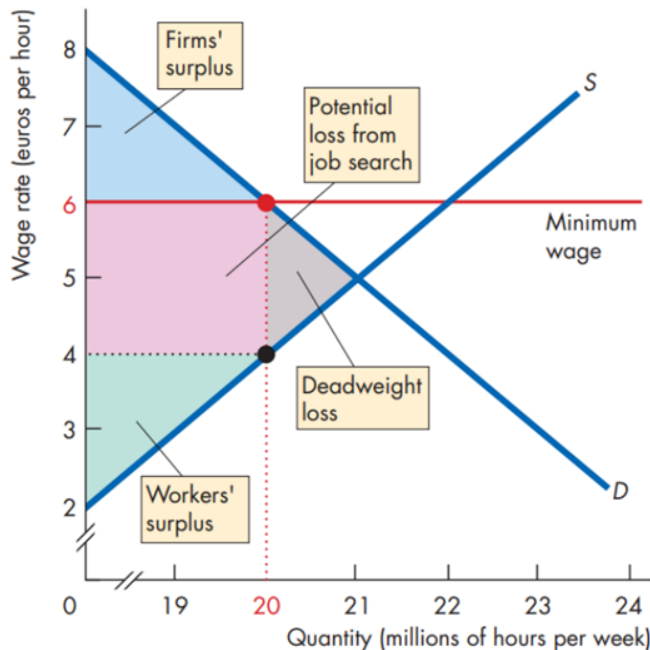
Workers' surplus shrinks

Deadweight loss (grey triangle) arises

Increased job search activity is represented by the red rectangle – borne by workers

Full loss from the minimum wage equals the deadweight loss (grey) plus the cost of increased job search (red)

Figure 6.4 The Inefficiency of a Minimum Wage



Externalities

- An *external cost* is a cost borne not by the producers but borne by other people.
 - Producing electricity from burning coal leads to air pollution and acid rain.
 - Producing chemicals can cause pollution to air and water.
 - Pollution, health-related problems.
 - Noise pollution.
- An *external benefit* is a benefit that accrues to people other than the buyer of a good.
 - Cycling to work helps to reduce the level of pollution and congestion. Therefore other road users have quicker journey times and help to reduce the level of pollution.
 - Reduced risk of transferring disease. If you get vaccinated, the personal benefit is your own reduction in risk of disease. Other people in society are less likely to be infected because you are healthy.

Public Goods & Common Resources

- A *public good* is a good or service that is consumed simultaneously by everyone, even if they don't pay for it.
 - national defence, law enforcement, fire services, street lamp etc.
- *Common resources* are resources that no one owns and that everyone can use.
 - fish in the ocean, city parks, footpath etc.
- Free rider problem arises (underproduction)

Monopoly

A *monopoly* is a firm that has sole control of a market.

- A monopoly can earn large profit but it prevents the market from achieving an efficient use of resources.
- To achieve profit maximization, it produces less than the efficient quantity and raises its price.

Taxes

Taxes

- Everything we earn and everything we buy is taxed.
- Taxes are an important source of *government revenue*.
- In a free market, the amount that consumers pay for a good or a service and the amount received by the sellers is same.
- Taxes drive a wedge between this – consumers pay more, producers receive less. The difference between what consumers pay and what producers get is the total tax incidence.

Tax incidence is the division of the *burden* of a tax between buyers and sellers.

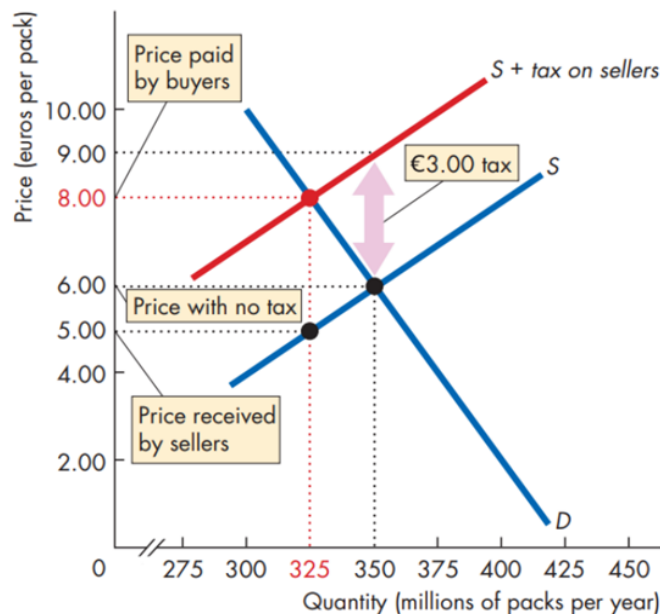
Tax incidence

- Tax incidence is determined by market forces.
- The division of the tax between buyers and sellers *depends on the elasticities of demand and supply*.
- In general, the more inelastic the demand for a good, the larger is the amount of the tax paid by buyers.
- In general, the more elastic the supply, the larger is the amount of the tax paid by buyers.

A tax on Sellers

- Let's assume a tax of 3 euros is imposed on cigarette sellers
- With no tax, equilibrium price is 6 euros per pack and quantity traded is 350 millions of packs per year
- A tax on sellers decreases supply
- To determine the position of the new supply curve, we add the tax to the minimum price that sellers are willing to accept for each quantity sold

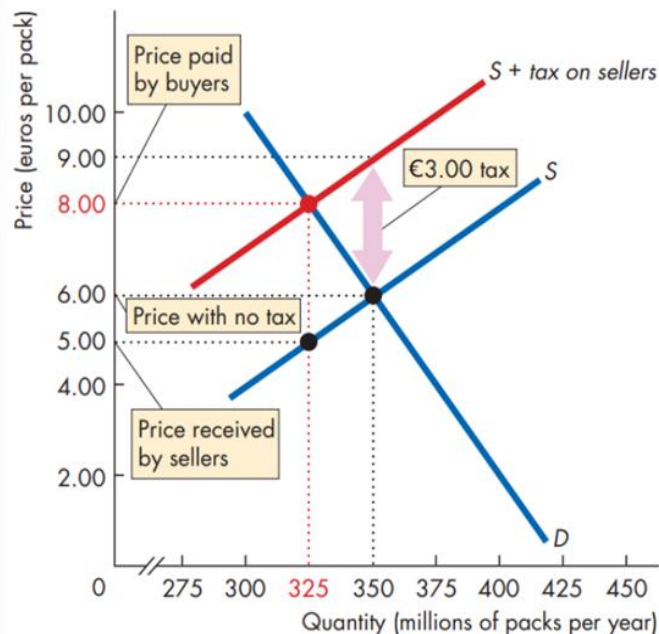
Figure 6.5 A Tax on Sellers



A tax on Sellers

- With the tax, they will sell 325 million packs only at a price of 8 euros
- The price paid by buyers rises by 2 euros to 8 euros a pack.
- The price received by sellers falls by 1 euro to 5 euros a pack (They collect 8 euros but give out 3 euros in tax, so end up with 5 euros)
- Tax incidence: Buyers pay 2 euros while sellers pay 1 euro of the tax.

Figure 6.5 A Tax on Sellers



A tax on Buyers

- Let's assume a sales tax of 3 euros is imposed on cigarette buyers
- A tax on buyers decreases demand
- To determine the position of the new demand curve, we deduct the tax from the maximum price that buyers are willing to pay for each quantity bought.
- With a 3 euros tax, they will buy 350 packs a year only if the price is 3 euros.

Figure 6.6 A Tax on Buyers



A tax on Buyers

- At the new equilibrium, 325 packs will be traded at a price of 5 euros per pack.
- The price received by sellers falls by 1 euro to 5 euros a pack
- The price paid by buyers increases by 2 euros to 8 euros (They pay 5 euros to the seller and 3 euros as tax)
- Tax incidence: Buyers pay 2 euros while sellers pay 1 euro of the tax.

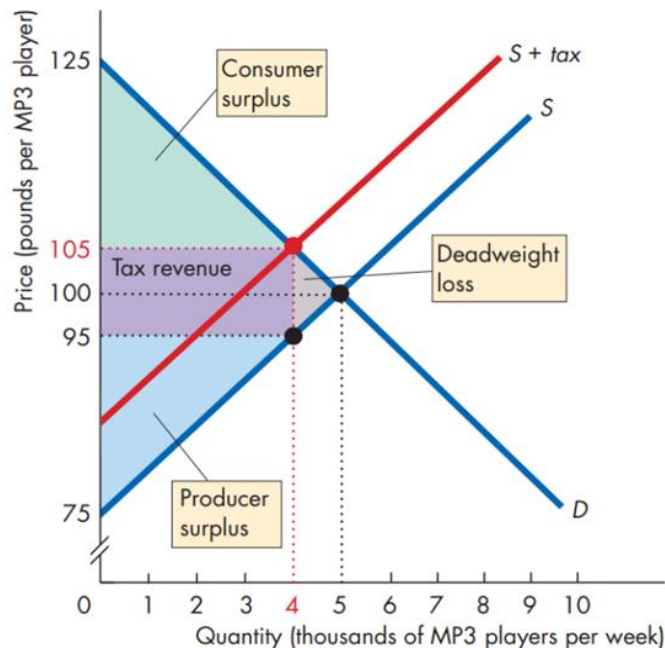
Figure 6.6 A Tax on Buyers



Taxes & Inefficiency

- Consider a market for MP3 players
- With no tax (free-market), the market produces an efficient quantity of 5000 MP3 players a week at a price of 100 pounds per player
- Suppose the government imposes a tax of £10 on sellers of DVD players and the tax burden is shared equally between producers and consumers
- With the tax, the supply curve shifts to the $S + \text{tax}$ (red) curve.
- This new supply curve does not show marginal social cost. It is only a transfer of resources to the government.
- Equilibrium quantity decreases to 4000 MP3 players a week
- Buyers pay £105 but sellers receive £95
- Both consumer and producer surplus shrinks
- Part of the loss of consumer surplus and producer surplus goes to the government as tax revenue (the purple area) and part becomes a deadweight loss (the grey area).

Figure 6.10 Taxes and Efficiency



Tax incidence

Try figuring out the four extreme cases:

Perfectly inelastic demand – buyers pay

Perfectly elastic demand – sellers pay

Perfectly inelastic supply – sellers pay

Perfectly elastic supply – buyers pay

Quotas and Subsidies

A production *quota* is an upper limit to the quantity of a good that may be produced in a specified period. A production quota imposed below the equilibrium quantity has the following effects:

1. Decreases supply of the good in the market
2. Increases price level

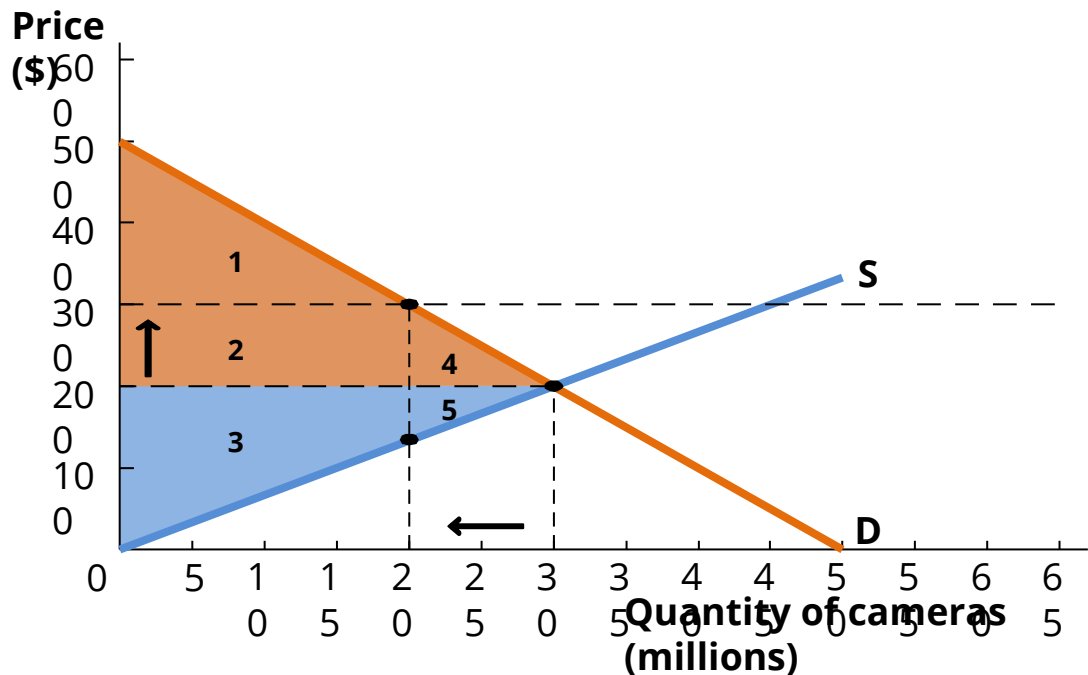
A production *subsidy* is a payment made by the government to a producer for each unit produced. A production subsidy has the following effects:

1. Increases supply of the good
2. A fall in price and increase in quantity traded

Practice Problems

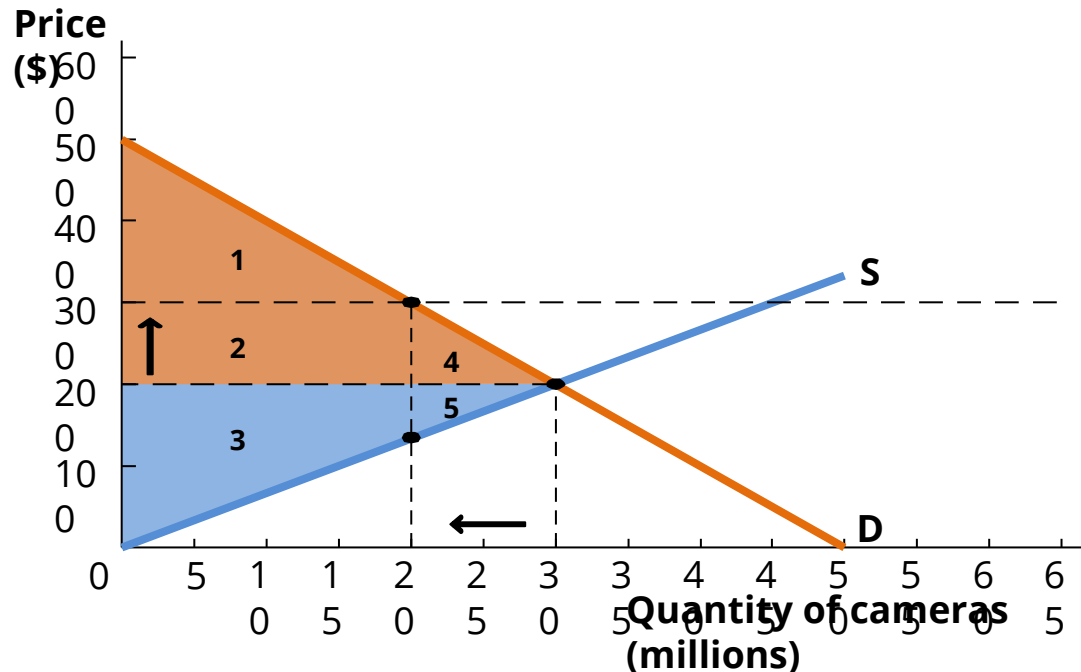
Active Learning: Calculating total surplus

Use the following graph to calculate the *difference* in total surplus if the price increases from \$200 to \$300.



Active Learning: Calculating total surplus

Use the following graph to calculate the *difference* in total surplus if the price increases from \$200 to \$300.



- When price is \$200
 - CS = \$4.5 B and PS = \$3B
 - Total surplus = \$7.5B
- When price is \$300
 - $CS = \frac{1}{2} \times 20M \times (\$500 - \$300) = \$2B$
 - $PS = \$3B - \frac{1}{2} \times 10M \times (\$200 - \$133.33) + 20M \times (\$300 - \$200) = \$4.33B$
 - Total surplus = \$2B + \$4.33B = \$6.33B
- The change in total surplus is equal to \$6.33B - \$7.5B = -\$1.17B

Market Efficiency

Demand for a good X is given by:

$$P = 2000 - 50Q$$

Supply for good X is given by:

$$P = 500 + 10Q$$

1. Find the equilibrium price and quantity
2. On a graph, draw the demand and supply curve and show the equilibrium price and quantity
3. Calculate consumer surplus & producer surplus
4. Calculate the total economic surplus
5. What is the efficient quantity of good X? Why?

Tax revenue

Consider the market for Good X.

Demand: $P = 80 - 2Q$ and Supply: $P = 40 + 2Q$

- a) Given the information, find the market equilibrium price and quantity for good X
- b) Calculate the consumer surplus, the producer surplus and the total surplus for good X

Suppose the government decides to impose a tax of TK 10 per unit on good X which is shared equally between buyers and sellers

- c) Given the tax, find the new price that buyers will pay and the new price that sellers will receive
- d) Find the new CS, PS and TS
- e) Calculate the tax revenue generated by the government and the deadweight loss from the tax

Summary

- *Key concepts:*
 - Willingness to pay
 - Marginal cost and marginal benefit
 - Consumer surplus
 - Producer surplus
 - Total surplus
 - Deadweight loss
 - Sources of market failures
 - Tax revenue
- *Reading: Chapter 5 of “Economics” by Parkin, Powell, Mathews 6th Ed.*